arXiv:2309.02047v1 [math.CV] 5 Sep 2023

ERDŐS–LAX ESTIMATES FOR RATIONAL POWERS OF POLYNOMIALS

ALEX BERGMAN & OLOF RUBIN

ABSTRACT. The classical Erdős–Lax inequality states that if P is a polynomial of degree n which is zero free on $\{|z| < 1\}$ then $\|P'\| \leq \frac{n}{2}\|P\|$ where the norm is taken as supremum norm on the unit circle. Equality holds if and only if all zeros of P are situated on the unit circle. In the present article we generalise the equality to functions of the form $f = P^{1/m}$ where P has all its roots on the unit circle obtaining $\|f'\| = \frac{n/m}{2}\|f\|$. We further prove an inequality of the form $\|f'\| \leq \frac{n/m}{2}\|f\|$ where P is zero free on $\{|z| < 1\}$.

1. INTRODUCTION

The purpose of this article is to establish estimates in the uniform norm of derivatives of fractional powers of polynomials. In particular, we consider functions $f = P^{1/m}$ where P is a polynomial and $m \in \mathbb{N}$ with the goal of relating the supremum norms of f' and f relative to the unit circle. Throughout the text we adopt the notation that $\|\cdot\|$ denotes the supremum norm on the unit circle, $\mathbb{T} = \{z : |z| = 1\}$. We further adopt the notation $\mathbb{D} = \{z : |z| < 1\}$ for the unit disk.

For $m > 1$ the function f will be multi-valued. However, f and its derivatives will have single-valued absolute values.

A classical example of a relation of the aforementioned type comes from Bernstein's inequality. If P is a polynomial of degree n then

$$\|P'\| \leq n\|P\| \tag{1}$$

with equality if and only if $P(z) = cz^n$ for some constant $c \in \mathbb{C}$.

Erdős suggested that this inequality can be sharpened in the case when P is zero free on $\mathbb{D}$. He suggested that a sharp inequality in this case is given by

$$\|P'\| \leq \frac{n}{2}\|P\|. \tag{2}$$

Date: September 6, 2023.

2020 *Mathematics Subject Classification.* 30A10. 26D05. 41A17. 30C10.

Key words and phrases. Polynomial inequalities, Erdős–Lax inequality, Fractional powers of polynomials.

This was later proven to be correct by Lax [6] using techniques developed by Szegő and Pólya. Szegő and Pólya verified the special case of equality in (2) when the zeros of P are situated on $\mathbb{T}$.

Lax was able to lift this result using so-called reciprocal polynomials to prove (2). We should mention that preceding Lax's proof of (2), Turán [10] showed that if P is a polynomial which is zero free on $\mathbb{C} \setminus \mathbb{D}$ then

$$\|P'\| < \frac{n}{2}\|P\|. \tag{3}$$

It is interesting to ask if the case of equality in (2) that was proven by Szegő and Pólya can be generalised to the setting where we instead of the polynomial P consider the function $f = P^{1/m}$. In particular, can we then replace the degree n in the equation with the number n/m? Here we provide the affirmative answer to this question. We also consider the case where the zeros are situated exterior to $\mathbb{D}$ and provide a case for which the inequality is valid. Several generalisations of the Erdős–Lax inequality exists for the case of polynomials. See, for example, [7, 8, 2]. In particular, we mention the L^p versions of the Erdős–Lax inequality proved in [9] ($0 < p < \infty$) which were obtained as consequences of L^p Bernstein estimates established by Arestov in [1]. We remark that the results were obtained earlier in the case $1 \leq p < \infty$. For such p the Berstein estimate was given by Zygmund [11] and the Erdős–Lax estimate was given by de-Bruijn [4]. However, the direction toward similar inequalities for fractional powers of polynomials appear to be unexplored.

As k ranges between $1, \dots, N$, let a_k be complex numbers, $|a_k| \geq 1$, s_k positive rational numbers, and $m \geq 1$ an integer such that $ms_k \in \mathbb{N}$ for all s_k. Denote by P the polynomial

$$P(z) = c^m \prod_{k=1}^{N} (z - a_k)^{s_k m}.$$

The degree of P will be denoted by $n = m \sum_{k=1}^{N} s_k$. Let

$$f(z) = P(z)^{1/m} = c \prod_{k=1}^{N} (z - a_k)^{s_k}, \quad z \in \mathbb{D} \tag{4}$$

where the branches are chosen as to make f analytic in $\mathbb{D}$. The function f extends continuously to the unit circle and the derivative of f is given by

$$f'(z) = \frac{1}{m} f(z) \frac{P'(z)}{P(z)} = \frac{f(z)}{m} \sum_{k=1}^{n} \frac{1}{z - c_k}, \tag{5}$$

where the c_k's are the zeros of P counting multiplicities. If $s_k \geq 1$ for all k, the derivative is also continuous in $\overline{\mathbb{D}}$. Our goal is to prove the following Erdős–Lax type equality.

Theorem 1. *Let $|a_k| = 1$ and let $s_k \geq 1$ denote rational numbers. If*

$$f(z) = c\prod_k (z - a_k)^{s_k},$$

then

$$\|f'\| = \frac{\sum s_k}{2}\|f\|.$$

Using this equality we are able to prove an Erdős–Lax type inequality with certain restrictions on the zeros of f in $\{z : |z| > 1\}$.

Theorem 2. *Let $|a_k| \geq 1$ and let $s_k \geq 1$ denote rational numbers. If*

$$f(z) = c\prod_k (z - a_k)^{s_k},$$

and $s_k \in \mathbb{N}$ whenever $|a_k| > 1$, then

$$\|f'\| \leq \frac{\sum s_k}{2}\|f\|$$

with equality if and only if all a_k's are unimodular.

This result allows us to have zeros of f away from the unit circle. However, the restriction is that the zeros which are located away from the circle must have integer exponents.

It should be noted that all our results extend to the case where s_k is real-valued with the same bound as specified above by a density argument.

The generalisation of the Erdős–Lax inequality presented here serves the purpose of allowing for fractional powers of roots on the unit circle. The motivation for this comes from applications to weighted Chebyshev problems on the unit circle. Given a weight function $w : \mathbb{T} \to [0, \infty)$ this amounts to finding the unique monic minimiser $T_n^w(z) = z^n + \sum_{k=0}^{n-1} a_k z^k$ such that

$$\|wT_n^w\| = \min_{b_k} \left\| w(z)\left(z^n + \sum_{k=0}^{n-1} b_k z^k\right)\right\|.$$

With Theorem 1 at hand it is possible to generalise work of Lachance, Saff and Varga [5] characterising such polynomials in the case where $w(z) = |z + 1|^s$ for $s > 0$. The reader is referred to [3] for more details.

2. Proofs of the main Theorems

Assume, as before, that f is of the form

$$f(z) = c\prod_k (z - a_k)^{s_k}$$

where $|a_k|, s_k \geq 1$. First of all, note that the restriction $s_k \geq 1$ is necessary for otherwise may f' not be bounded on $\mathbb{T}$ if $|a_k| = 1$ for some k. Secondly, note that if $f(z) = (z-a)^s$ with $|a| \geq 1$ and $s \geq 1$ then

$$f'(z) = s(z-a)^{s-1},$$

and thus $\|f'\| = s(2+(|a|-1))^{s-1}$. Since $\|f\| = (2+(|a|-1))^s$ this implies that

$$\frac{\|f'\|}{\|f\|} = \frac{s}{(2+|a|-1)} \leq \frac{s}{2}$$

with equality if and only if $|a| = 1$. Also note that in this case f' achieves its maximum modulus at the same point on the unit circle as f does. This is something we will investigate further. The argument we now give will be an adaptation of Szegő's argument for polynomials suitably modified to our situation. The key step is to prove that any point of maximality for f' is a point of maximality for f (with one exception).

Proof of Theorem 1. Recall that we are assuming $s_k \geq 1$, for all k and that $\sum_{k=1}^N s_k = n/m$ where n is the degree of the polynomial P satisfying $f(z) = P(z)^{1/m}$. With these restrictions on s_k, we clearly see that f' is continuous up to the circle. The second derivative of f is given by

$$\begin{aligned} f''(z) &= \frac{1}{m}\left(f'(z)\frac{P'(z)}{P(z)} + f(z)\frac{P''(z)P(z)-(P'(z))^2}{P(z)^2}\right) \\ &= \frac{1}{m}\left(f'(z)\frac{P'(z)}{P(z)} + mf'(z)\frac{P(z)}{P'(z)}\frac{P''(z)P(z)-(P'(z))^2}{P(z)^2}\right) \\ &= \frac{f'(z)}{m}\left((1-m)\frac{P'(z)}{P(z)} + m\frac{P''(z)}{P'(z)}\right). \end{aligned}$$

With the intent of initially proving that $\|f'\| \leq \frac{n/m}{2}\|f\|$, we let $z_0 \in \mathbb{T}$ be a maximum of f'. Consider first the case where $P(z_0) \neq 0$ and hence also $P'(z_0) \neq 0$ (otherwise $f'(z_0) = 0$). The derivative of the map $\theta \mapsto |f'(z_0 e^{i\theta})|^2$ attains the value 0 at $\theta = 0$. Thus

$$z_0 f''(z_0)\overline{f'(z_0)} = \overline{z_0} f'(z_0)\overline{f''(z_0)},$$

meaning that $z_0 f''(z_0)/f'(z_0) \in \mathbb{R}$. Combining this with the relation for f'', we obtain

$$\mathbb{R} \ni z_0 \cdot \frac{f''(z_0)}{f'(z_0)} = \frac{1}{m}\left((1-m)\mathrm{Re}\left(z_0\frac{P'(z_0)}{P(z_0)}\right) + m\mathrm{Re}\left(z_0\frac{P''(z_0)}{P'(z_0)}\right)\right). \tag{6}$$

For $z \in \mathbb{T}$,

$$z \cdot \frac{P'(z)}{P(z)} = \sum_{k=1}^n \frac{z}{z-a_k} = \frac{n}{2} + i\sum_{k=1}^n t_k(z),$$

where $t_k(z) = \mathrm{Im}\Big(z(z-a_k)^{-1}\Big)$. If $\sum_{k=1}^n t_k(z_0) = 0$ then we attain from equation (5) the inequality

$$\|f'\| = |f'(z_0)| = \frac{n/m}{2}|f(z_0)| \leq \frac{n/m}{2}\|f\|.$$

We want to show that this is the only possible case (with one exception) and hence we assume that $\sum_{k=1}^n t_k(z_0) \neq 0$. We consider the quotients

$$z^2 \cdot \frac{P''(z)}{P(z)} = \left(\sum_{k=1}^n \frac{z}{z-a_k}\right)^2 - \sum_{k=1}^n \left(\frac{z}{z-a_k}\right)^2,$$

$$z \cdot \frac{P''(z)}{P'(z)} = z^2\frac{P''(z)}{P(z)} / z\frac{P'(z)}{P(z)} = \sum_{k=1}^n \frac{z}{z-a_k} - \frac{\sum_{k=1}^n \left(\frac{z}{z-a_k}\right)^2}{\sum_{k=1}^n \frac{z}{z-a_k}}.$$

From (6) we see

$$\begin{aligned}&(1-m)\sum_{k=1}^n t_k(z_0)\\ &+m\left(\sum_{k=1}^n t_k(z_0) - \frac{n/2\sum_{k=1}^n t_k(z_0) - \sum_{k=1}^n t_k(z_0)\sum_{k=1}^n (1/4 - t_k(z_0)^2)}{n^2/4 + \left(\sum_{k=1}^n t_k(z_0)\right)^2}\right) = 0.\end{aligned}$$

Thus we obtain

$$\frac{n^2}{4} + \left(\sum_{k=1}^n t_k(z_0)\right)^2 = m\left(n/2 - \sum_{k=1}^n (1/4 - t_k(z_0)^2)\right).$$

Substituting this into the equation for $\mathrm{Re}(z_0P''(z_0)/P'(z_0))$, we obtain

$$\begin{aligned}\mathrm{Re}\left(z_0\frac{P''(z_0)}{P'(z_0)}\right) &= \frac{n}{2} - \frac{n/2\sum_{k=1}^n (1/4 - t_k(z_0)^2) + \left(\sum_{k=1}^n t_k(z_0)\right)^2}{n^2/4 + \left(\sum_{k=1}^n t_k(z_0)\right)^2}\\ &= \frac{n}{2} - \frac{n/2\left(n/2 - n^2/4m - \left(\sum_{k=1}^n t_k(z_0)\right)^2/m\right) + \left(\sum_{k=1}^n t_k(z_0)\right)^2}{n^2/4 + \left(\sum_{k=1}^n t_k(z_0)\right)^2}\\ = \frac{n}{2} - \frac{(1-n/2m)n^2/4 + (1-n/2m)\left(\sum_{k=1}^n t_k(z_0)\right)^2}{n^2/4 + \left(\sum_{k=1}^n t_k(z_0)\right)^2} &= \frac{n}{2}\left(1 + \frac{1}{m}\right) - 1.\end{aligned}$$

Finally, we obtain that

$$z_0\frac{f''(z_0)}{f'(z_0)} = m^{-1}\left((1-m)\frac{n}{2} + m\left(\frac{n}{2}\left(1+\frac{1}{m}\right) - 1\right)\right) = \frac{n}{m} - 1,$$

whenever z_0 is a maximum of f' such that $\sum t_k(z_0) \neq 0$. It is easily seen that $(f')^m = Q$ is a polynomial of degree $n-m$. Thus

$$\frac{z_0}{m}\frac{Q'(z_0)}{Q(z_0)} = z_0\frac{f''(z_0)}{f'(z_0)} = \frac{n}{m} - 1.$$

Since Q attains its maximum modulus at the same point, z_0, as f', it follows from the condition for equality in Bernstein's inequality that $f' = cz^{n/m-1}$ (also, n/m must be an integer). In this case one verifies by direct computation that the inequality holds. Otherwise,

$$\sum_{k=1}^{n} t_k(z_0) = 0.$$

In conclusion we obtain the desired inequality

$$\|f'\| = |f'(z_0)| = \frac{n/m}{2}|f(z_0)| \leq \frac{n/m}{2}\|f\|.$$

It remains to consider the case when $P(z_0) = 0$. Then $z_0 = a_k$ for some k and $s_k = 1$, since otherwise $f'(z_0) = 0$. Then the right-hand side of

$$\frac{f''(z)}{f'(z)} = \frac{1}{m}\left((1-m)\frac{P'(z)}{P(z)} + m\frac{P''(z)}{P'(z)}\right),$$

has a singularity at a_k (unless $m = 1$, in which case it is nothing but the classical version of the Theorem), but the left-hand side does not since f'' is continuous at a_k and f' has a maximum. Hence this cannot happen (unless $f' \equiv 0$).

The lower bound $\|f'\| \geq \frac{n/m}{2}\|f\|$ follows from Lemma 1 which generalises Turán's inequality. □

Lemma 1. *Let $f(z) = c\prod_{k=1}^{N}(z-a_k)^{s_k}$ with $s_k \geq 1$ and $|a_k| \leq 1$. Then*

$$\|f'\| \geq \frac{\sum s_k}{2}\|f\|.$$

Proof. By differentiating, we obtain

$$f'(z) = f(z)\sum_{k=1}^{N}\frac{s_k}{z-a_k}$$

and the result follows if we can show that

$$\left|\sum_{k=1}^{N}\frac{s_k}{z-a_k}\right| \geq \frac{\sum s_k}{2} \quad \text{for } z \in \mathbb{T}.$$

To see this, note that, with $|z| = 1$,

$$\begin{aligned}\operatorname{Re}\sum_{k=1}^{N} s_k \frac{z}{z-a_k} &= \frac{1}{2}\sum_{k=1}^{N} s_k \left(\frac{z}{z-a_k} + \frac{\overline{z}}{\overline{z-a_k}}\right) \\ &= \frac{1}{2}\sum_{k=1}^{N} s_k \frac{z(\overline{z}-\overline{a_k}) + \overline{z}(z-a_k)}{|z-a_k|^2} \\ &= \frac{1}{2}\sum_{k=1}^{N} s_k \frac{2-2\operatorname{Re}(z\overline{a_k})}{1-2\operatorname{Re}(z\overline{a_k})+|a_k|^2} \geq \frac{1}{2}\sum_{k=1}^{N} s_k.\end{aligned}$$

□

From the proof of Theorem 1 we gather the following corollary.

Corollary 1. *Let*

$$f(z) = c\prod_{k=1}^{N}(z-a_k)^{s_k}$$

with $|a_k| = 1$ and $s_k \geq 1$. Then f' is maximal on the unit circle at any point z for which $|f(z)| = \|f\|$ and in that case

$$\left|\sum_{k=1}^{N}\frac{s_k}{z-a_k}\right| = \operatorname{Re}\sum_{k=1}^{N} s_k\frac{z}{z-a_k} = \frac{\sum_{k=1}^{N} s_k}{2}.$$

Even though it is not difficult to conclude this result for polynomials using Lax's argument, it appears to the best of our knowledge that this observation is new, even for the case where f is a polynomial.

We now proceed to consider the case where $|a_k| \geq 1$.

Proof of Theorem 2. Let $f^*(z)$ denote the reciprocal of f defined via

$$f^*(z) = \overline{c}\prod_{k}(1-\overline{a_k}z)^{s_k}.$$

The reciprocal can equivalently be defined as an mth root of the reciprocal of P. Indeed, in this case

$$P(z) = f(z)^m = c^m\prod_{k=1}^{N}(z-a_k)^{ms_k}$$

whence

$$P^*(z) = \overline{c}^m\prod_{k=1}^{N}(1-\overline{a_k}z)^{ms_k}.$$

We see that $f^*(z) = P^*(z)^{1/m}$. Now it is easily verified that for $z \in \mathbb{T}$, $|P(z)| = |P^*(z)|$ and $|P'(z)| \leq |(P^*)'(z)|$. The last inequality holds since we are assuming

that all the zeros of P are exterior to $\mathbb{D}$. But this implies that for $z \in \mathbb{T}$, we have $|f(z)| = |f^*(z)|$ and

$$|(f^*)'(z)| = \left|\frac{1}{m}P^*(z)^{1/m-1}(P^*)'(z)\right| \geq \left|\frac{1}{m}P^{1/m-1}(z)P'(z)\right| = |f'(z)|.$$

Note further that under the above assumptions,

$$f^*(z) = \prod_j (z - a_{k_j})^{s_{k_j}} Q(z)$$

with $|a_{k_j}| = 1$ where the indices k_j correspond to those s_{k_j} which are non-integer and Q is a polynomial all of whose zeros are inside $\mathbb{D}$. We conclude that for any $|\zeta| = 1$, it holds true that

$$f(z) + \zeta f^*(z) = \prod_j (z - a_{k_j})^{s_{k_j}} \tilde{Q}(z),$$

where $\tilde{Q}$ is some polynomial which has all its zeros on the unit circle. To see this, note that

$$\log\left|\frac{f^*}{f}\right|$$

is subharmonic on $\mathbb{D}$ and superharmonic on $\mathbb{C} \setminus \mathbb{D}$ while constantly equal to 0 on the unit circle. Hence $f+\zeta f^*$ has all its zeros on $\mathbb{T}$, the same also being true for $\tilde{Q}$. Therefore we can apply Theorem 1 to the function $g = f + \zeta f^*$ where ζ is chosen such that $\arg f'(z_0) = \arg \zeta (f^*)'(z_0)$ for a point z_0 satisfying $|f'(z_0)| = \|f'\|$. But then

$$\|g'\| \geq |g'(z_0)| = |f'(z_0)| + |(f^*)'(z_0)| \geq 2|f'(z_0)| = 2\|f'\|.$$

Since we further have that

$$\|g'\| = \frac{\sum s_k}{2}\|g\| \leq \left(\sum s_k\right)\|f\|$$

the result follows.

□

3. Concluding remarks

Theorem 1 is a direct generalisation of a result due to Szegő and Pólya that first appeared in print in [6]. Theorem 2 generalises the Erdős–Lax inequality. However, we believe that this result should be valid without restrictions on the zeros in the exterior of the closed unit disc. In fact, we conjecture the following.

Conjecture 1. *Let $P(z) = c\prod_{k=1}^N (z - a_k)^{ms_k}$ where $c \in \mathbb{C}$, $m \in \mathbb{N}$, $s_k \geq 1$, and $|a_k| \geq 1$. Then for any branch $f(z) = P(z)^{1/m}$, we have*

$$\|f'\| \leq \frac{\sum_{k=1}^N s_k}{2}\|f\|$$

with equality if and only if $|a_k| = 1$ for all k.

The reason that the techniques developed in this article does not seem to work to prove this case is that for such functions f it is not guaranteed that $f + \zeta f^*$ from the proof of Theorem 2 is a fractional power of a polynomial. Instead, we believe that new ideas are needed to prove this conjecture.

References

[1] Vitalii Vladimirovich Arestov. On integral inequalities for trigonometric polynomials and their derivatives. *Izvestiya Rossiiskoi Akademii Nauk. Seriya Matematicheskaya*, 45(1):3–22, 1981.

[2] A. Aziz and Q. G. Mohammad. Simple proof of a theorem of Erdős and Lax. *Proceedings of the American Mathematical Society*, 80(1):119–122, 1980.

[3] J. S. Christiansen and O. Rubin. Chebyshev polynomials on lemniscates. *to appear*, 2023.

[4] N. G. de Bruijn. Inequalities concerning polynomials in the complex domain. *Nederl. Akad. Wetensch., Proc.*, 50:1265–1272 = Indagationes Math. **9, 591–598**, 1947.

[5] M. Lachance, E. B. Saff, and R. S. Varga. Inequalities for polynomials with a prescribed zero. *Mathematische Zeitschrift*, 168(2):105–116, Jun 1979.

[6] P. Lax. Proof of a conjecture of P. Erdős on the derivative of a polynomial. *Bulletin of the American Mathematical Society*, 50(8):509 – 513, 1944.

[7] M. A. Malik. On the Derivative of a Polynomial. *Journal of the London Mathematical Society*, s2-1(1):57–60, 01 1969.

[8] A. Mir, A. Hussain, and F. A. Wagay. On the erdös-lax inequality and its various generalizations concerning polynomials. *Rendiconti del Circolo Matematico di Palermo Series 2*, 72(3):2025–2037, Apr 2023.

[9] Q.I Rahman and G Schmeisser. L^p inequalities for polynomials. *Journal of Approximation Theory*, 53(1):26–32, 1988.

[10] P. Turán'. über die ableitung von polynomen. *Compositio Math*, 7:89–95, 1939.

[11] A. Zygmund. A Remark on Conjugate Series. *Proc. London Math. Soc. (2)*, 34(5):392–400, 1932.

Centre for Mathematical Sciences, Lund University, Box 118, SE-22100, Lund, Sweden,

Email address: alex.bergman@math.lu.se, olof.rubin@math.lth.se